

Week 6 Assignment Report

Git, GitHub & Final Data Analytics Project

[gdrive_link](#)

[github_link](#)

Name: VISHAL

Student ID: INST26DA030224

Email: vk4499573@gmail.com

Internship Domain: Data Analytics

Batch Cohort: Batch-03

Git, GitHub Tasks

Task 1:

Solution:

1. Let's create new a repository named 'InternNova-Final'.

The screenshot shows the GitHub 'Create a new repository' page. The browser address bar shows 'github.com/new'. The page title is 'Create a new repository'. Below the title, it says 'Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository.](#) Required fields are marked with an asterisk (*)'. The page is divided into two sections: '1 General' and '2 Configuration'.

1 General

Owner * vishalbairwapyd / Repository name * InternNova-Final
InternNova-Final is available.

Great repository names are short and memorable. How about [verbose-doodle?](#)

Description Final week 6 task
17 / 350 characters

2 Configuration

Choose visibility * Public
Choose who can see and commit to this repository

Add README Off
READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore No .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)

Add license No license
Licenses explain how others can use your code. [About licenses](#)

[Create repository](#)

0.3 KB/s 0.4 KB

- Let's add a 'task1.txt' file, link remote repository and push the local changes on remote.

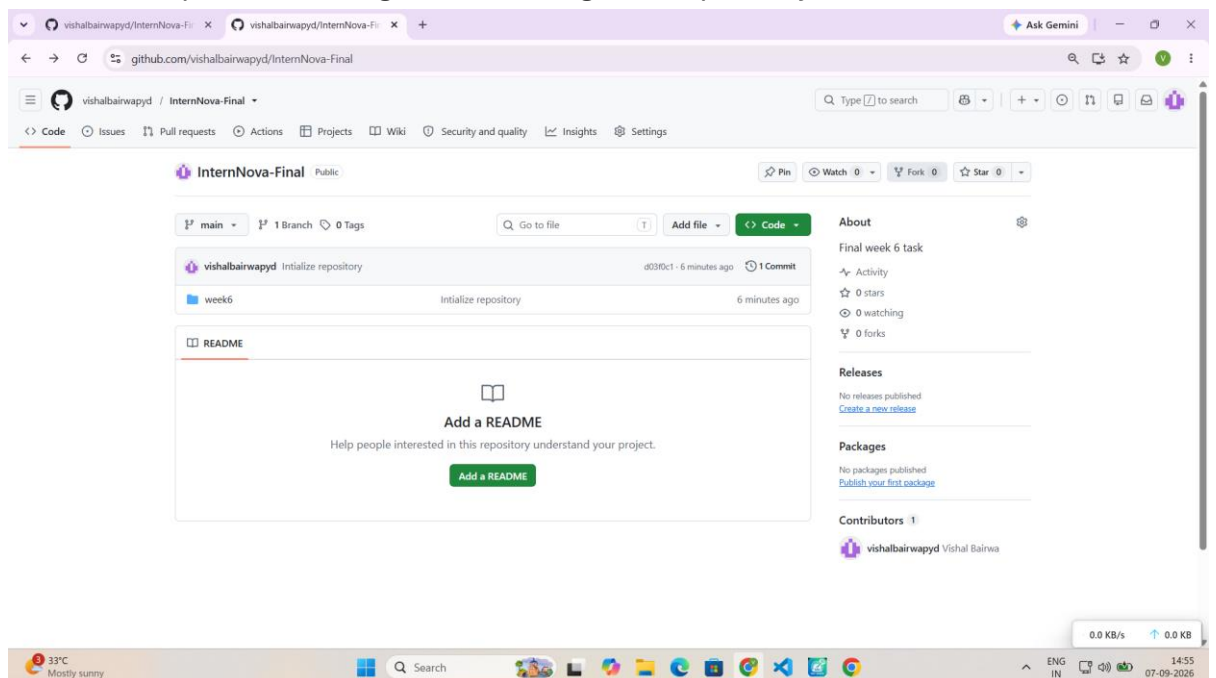
```
PS G:\Internship Final> git init
Initialized empty Git repository in G:\Internship Final\.git/
PS G:\Internship Final> git remote add origin https://github.com/vishalbairwapyd/InternNova-Final.git
PS G:\Internship Final> git add .
PS G:\Internship Final> git commit -m "Initialize repository"
[main (root-commit) d03f0c1] Initialize repository
1 file changed, 1 insertion(+)
create mode 100644 week6/task1.txt
PS G:\Internship Final> git push
fatal: The current branch main has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin main

To have this happen automatically for branches without a tracking
upstream, see 'push.autoSetupRemote' in 'git help config'.

PS G:\Internship Final> git branch -M main
PS G:\Internship Final> git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 316.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/vishalbairwapyd/InternNova-Final.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
PS G:\Internship Final>
```

- In the below screenshot you can see, we have successfully completed our first commit and push the changes on remote github repository.



Task 3:

Solution:

- **Git add, commit, and push**

```
PS G:\Internship Final> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   week6/task5.pbix

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        week6/task8.pptx

no changes added to commit (use "git add" and/or "git commit -a")
PS G:\Internship Final> git add .
PS G:\Internship Final> git commit -m "PPT added"
[main affdba6] PPT added
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 week6/task8.pptx
PS G:\Internship Final> git push
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 807.53 KiB | 26.05 MiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/vishalbairwapyd/InternNova-Final.git
   be102aa..affdba6  main -> main
```

- **Git commits**

```

commit affdba65586b276d3abff589a795c849f71b32d9 (HEAD -> main, origin/main)
Author: Vishal Bairwa <vishalbairwapyd@gmail.com>
Date: Thu Sep 10 11:53:13 2026 +0530

    PPT added

commit be102aa1af71a3e8d76af9b5dc7d7716c7f7b87e
Author: Vishal Bairwa <vishalbairwapyd@gmail.com>
Date: Wed Sep 9 17:09:32 2026 +0530

    Add readme

commit 71a8ca47c9f2dac047de1d42e5ad021d56bfed1f
Author: Vishal Bairwa <vishalbairwapyd@gmail.com>
Date: Wed Sep 9 17:07:44 2026 +0530

    Task 5, 6, 7 has completed

commit 27e09b59ea1e44874fe875ff2771d7fc0ed148ed
Author: Vishal Bairwa <vishalbairwapyd@gmail.com>
Date: Wed Sep 9 15:06:02 2026 +0530

    Task4 EDA has completed

commit db717d63c287f87ad17702b820b5fe6d0586a03a
Author: Vishal Bairwa <vishalbairwapyd@gmail.com>
Date: Tue Sep 8 23:12:53 2026 +0530

    Task 3 completed

commit d03f0c117e03e47a953cc24c894de4e1cd3799fc
Author: Vishal Bairwa <vishalbairwapyd@gmail.com>
Date: Mon Sep 7 14:48:25 2026 +0530

    Intialize repository
PS G:\Internship Final>

```

- **Git Pull**

```

PS G:\Internship Final> git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS G:\Internship Final> git pull
Already up to date.
PS G:\Internship Final>

```

- **Git Clone**

```
PS C:\Users\vk449\Downloads\clone_repo> git clone https://github.com/vis  
halbairwapyd/InternNova-Final  
Cloning into 'InternNova-Final'...  
remote: Enumerating objects: 28, done.  
remote: Counting objects: 100% (28/28), done.  
remote: Compressing objects: 100% (17/17), done.  
remote: Total 28 (delta 7), reused 26 (delta 5), pack-reused 0 (from 0)  
Receiving objects: 100% (28/28), 9.17 MiB | 1.95 MiB/s, done.  
Resolving deltas: 100% (7/7), done.  
PS C:\Users\vk449\Downloads\clone_repo> █
```

Task 3:

Solution:

Dataset Introduction: -

For this project, I have selected the **Superstore Sales Dataset**, which belongs to the **Retail and E-Commerce domain**.

The dataset contains information about customer orders, products, sales transactions, shipping, discounts, profit, customer segments, and geographical regions. It contains approximately **9, 994 records and 21 columns**, making it suitable for performing data preparation, exploratory data analysis, data visualization, and Power BI dashboard development.

The dataset includes important business fields such as **Order ID, Order Date, Customer Name, Customer Segment, Region, Product Name, Category, Sales, Quantity, Discount, and Profit**.

Main Areas Covered by the Dataset: -

- **Customers:** Customer name, customer segment, city, state, and region.
- **Orders:** Order ID, order date, shipping date, and shipping mode.
- **Products:** Product name, category, and sub-category.
- **Sales:** Sales amount and quantity sold.
- **Profitability:** Profit and discount information.
- **Geography:** Country, state, city, and region.

Untitled - Power BI Desktop

Search

File Home Help Table tools

Clipboard Copy Paste

Get data Excel OneLake SQL Server data

Enter data Datasource Recent sources

Transform Refresh data

Manage relationships Relationships

New Quick New New

measure measure column Calculations

table table

Manage roles View as

Security

Sensitivity

Share

Table: Superstore (9994 rows)

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category
43	CA-2016-101343	7/17/2016	7/22/2016	Standard Class	RA-19885	Ruben Ausman	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10003479	OFF-PA-10003479
514	CA-2017-163405	12/21/2017	12/25/2017	Standard Class	BN-11515	Bradley Nguyen	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10003811	OFF-PA-10003811
515	CA-2017-163405	12/21/2017	12/25/2017	Standard Class	BN-11515	Bradley Nguyen	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10001246	OFF-PA-10001246
1606	US-2016-115819	4/19/2016	4/24/2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10000823	OFF-PA-10000823
1607	US-2016-115819	4/19/2016	4/24/2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10000456	OFF-PA-10000456
1609	US-2016-115819	4/19/2016	4/24/2016	Second Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10002377	OFF-PA-10002377
1807	US-2016-116729	12/25/2016	12/28/2016	First Class	GK-14620	Grace Kelly	Corporate	United States	Los Angeles	California	90049	West	OFF-PA-10002005	OFF-PA-10002005
2060	CA-2014-106439	10/31/2014	11/4/2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-PA-10002975	OFF-PA-10002975
2061	CA-2014-106439	10/31/2014	11/4/2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10003996	OFF-PA-10003996
2063	CA-2014-106439	10/31/2014	11/4/2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-PA-10000477	OFF-PA-10000477
2065	CA-2014-106439	10/31/2014	11/4/2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10001963	OFF-PA-10001963
2068	CA-2014-106439	10/31/2014	11/4/2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-AR-10001419	OFF-PA-10001419
2191	CA-2016-118913	6/25/2016	6/29/2016	Standard Class	AS-10240	Alan Shonely	Consumer	United States	Los Angeles	California	90049	West	OFF-AP-10000692	OFF-PA-10000692
2492	CA-2014-113579	12/13/2014	12/15/2014	Second Class	KD-16345	Katherine Duch	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10001457	OFF-PA-10001457
2944	CA-2017-126242	11/19/2017	11/24/2017	Standard Class	MC-18100	Mick Crebagga	Consumer	United States	Los Angeles	California	90049	West	OFF-ST-10000675	OFF-PA-10000675
3534	CA-2014-101849	4/18/2014	4/23/2014	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10002375	OFF-PA-10002375
3535	CA-2014-101849	4/18/2014	4/23/2014	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	West	OFF-SU-10000657	OFF-PA-10000657
3536	CA-2014-101849	4/18/2014	4/23/2014	Standard Class	JL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10001034	OFF-PA-10001034
3565	CA-2016-130029	7/3/2016	7/6/2016	First Class	GT-14755	Guy Thornton	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10000552	OFF-PA-10000552
3566	CA-2016-130029	7/3/2016	7/6/2016	First Class	GT-14755	Guy Thornton	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10001135	OFF-PA-10001135
3759	CA-2015-167745	9/18/2015	9/23/2015	Standard Class	GB-14530	George Bell	Corporate	United States	Los Angeles	California	90049	West	OFF-AR-10003602	OFF-PA-10003602
3760	CA-2015-167745	9/18/2015	9/23/2015	Standard Class	GB-14530	George Bell	Corporate	United States	Los Angeles	California	90049	West	OFF-SU-10004782	OFF-PA-10004782
3844	CA-2014-101931	10/28/2014	10/31/2014	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	West	OFF-SU-10002301	OFF-PA-10002301
3845	CA-2014-101931	10/28/2014	10/31/2014	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	West	OFF-SU-10000646	OFF-PA-10000646
3847	CA-2014-101931	10/28/2014	10/31/2014	First Class	TS-21370	Todd Sumrall	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10003442	OFF-PA-10003442
4339	CA-2015-125234	11/27/2015	12/1/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Los Angeles	California	90049	West	OFF-PA-10000482	OFF-PA-10000482
4481	CA-2016-109365	11/3/2016	11/8/2016	Standard Class	XP-21865	Xylona Preis	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10001561	OFF-PA-10001561

0.0 KB/s 0.0 KB

32°C Mostly clear

Search

ENG IN 21:48 07-09-2026

Set Correct Data Types: -

Befor correction

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Recent Enter Source Sources Data

Data source settings Parameters Output Data

Export query results Refresh Preview Manage Query

Choose Remove Columns Columns Manage Columns

Keep Remove Rows Rows Sort

Split Column Group By

Replace Values Transform

Data Type: Whole Number

Merge Queries

Use First Row as Headers

Append Queries

Combine Files

Combine

Queries [1]

Superstore

Table.TransformColumnTypes(*Use First Row as Headers*,{"Row ID", Int64.Type, ("Order ID", type text), ("Order Date", type text), ("Ship Date", type text), ("Ship Mode", type text), ("Customer ID", type text), ("Customer Name", type text), ("Segment", type text), ("Country", type text), ("City", type text), ("State", type text), ("Postal Code", Int64.Type), ("Region", type text), ("Product ID", type text), ("Category", type text), ("Sub-Category", type text), ("Product Name", type text), ("Sales", type number), ("Quantity", Int64.Type), ("Discount", type number), ("Profit", type number)}])

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
2480	CA-2017-141992	6/19/2017	6/25/2017	Standard Class	FD-14305	Frank Olsen
2481	CA-2017-141992	6/19/2017	6/25/2017	Standard Class	FD-14305	Frank Olsen
2482	CA-2017-147956	12/24/2017	12/31/2017	Standard Class	AH-10210	Alan Hwang
2483	CA-2017-126067	8/28/2017	9/3/2017	Standard Class	KN-16705	Kristina Nunn
2484	CA-2017-126067	8/28/2017	9/3/2017	Standard Class	KN-16705	Kristina Nunn
2485	CA-2017-126067	8/28/2017	9/3/2017	Standard Class	KN-16705	Kristina Nunn
2486	CA-2017-126067	8/28/2017	9/3/2017	Standard Class	KN-16705	Kristina Nunn
2487	CA-2015-104514	1/2/2015	1/4/2015	Second Class	CB-12535	Claudia Berg
2488	CA-2015-104514	1/2/2015	1/4/2015	Second Class	CB-12535	Claudia Berg
2489	CA-2015-104514	1/2/2015	1/4/2015	Second Class	CB-12535	Claudia Berg
2490	CA-2015-141040	10/9/2015	10/13/2015	Second Class	TB-21250	Tim Brockma
2491	CA-2015-141040	10/9/2015	10/13/2015	Second Class	TB-21250	Tim Brockma
2492	CA-2014-113579	12/13/2014	12/15/2014	Second Class	KD-16345	Katherine Du
2493	CA-2014-144624	11/19/2014	11/23/2014	Standard Class	JM-15865	John Murray
2494	CA-2016-146206	9/10/2016	9/14/2016	Second Class	KT-16480	Kean Thornto
2495	CA-2016-146206	9/10/2016	9/14/2016	Second Class	KT-16480	Kean Thornto
2496	CA-2014-136644	6/16/2014	6/22/2014	Standard Class	SC-20575	Sonia Coolay

21 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

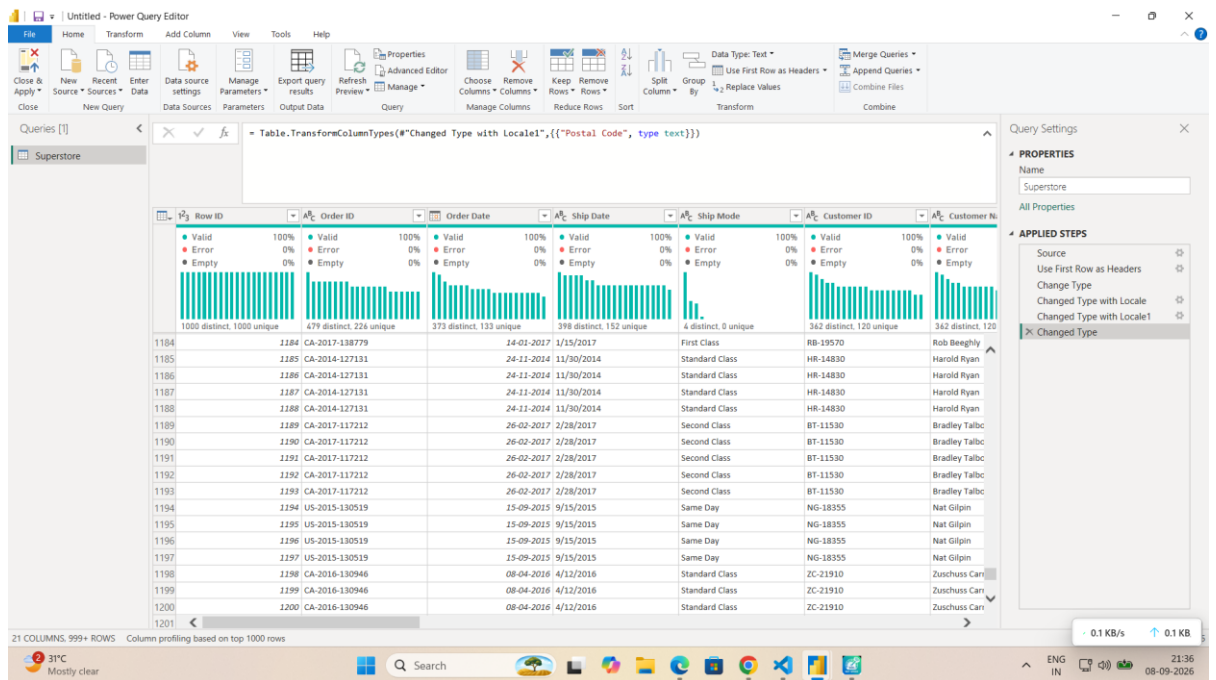
0.2 KB/s 0.1 KB

31°C Mostly clear

Search

ENG IN 21:29 08-09-2026

After Correction



Check Missing Values: -

Power Query's Column Quality and Column Profile features were used to identify missing.

The dataset contained:

- **100% valid values**
- **0% empty values**
- **0% errors**

Check Duplicate Records: -

In the below Screenshot you can see there are 5,009 distinct values out of 9994 rows means duplicate records can be there.

But we know that a order can contain multiple products, so the same Order ID can legitimately appear on several rows.

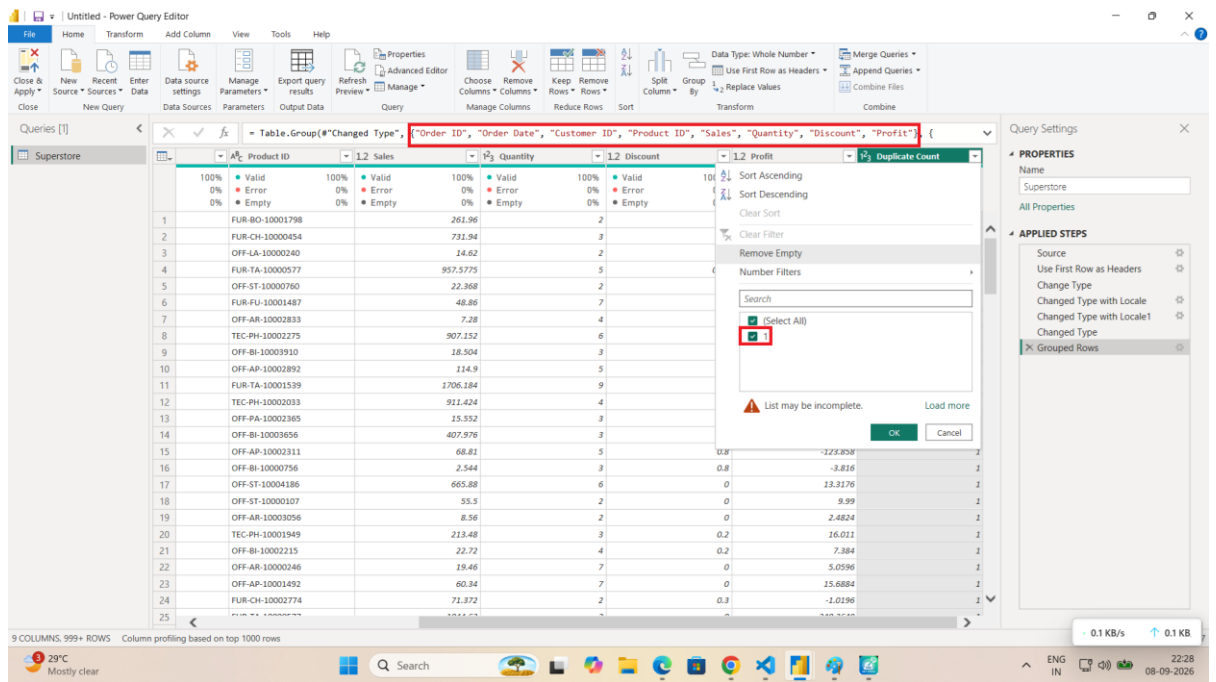
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Product Name	Quantity	Discount	Profit	Sales	Ship Date	Ship Mode	State	Sub-Category
2718	CA-2014-100006	07 September 2014	9/13/2014	Standard Class	DK-13375	Dennis Kane	Consumer	United States	New York Cr	New York	10024	East	TEC-PH-10002075									
6288	CA-2014-100096	08 July 2014	7/12/2014	Standard Class	EB-13705	Ed Braxton	Corporate	United States	San Francisco	California	94122	West	FUR-TA-10003715									
6289	CA-2014-100096	08 July 2014	7/12/2014	Standard Class	EB-13705	Ed Braxton	Corporate	United States	San Francisco	California	94122	West	OFF-BI-10001597									
9515	CA-2014-100293	14 March 2014	3/18/2014	Standard Class	NF-18475	Neil Franzöisch	Home Office	United States	Jacksonville	Florida	32216	South	OFF-PA-10000176									
3084	CA-2014-100328	28 January 2014	2/3/2014	Standard Class	JC-15340	Jasper Cacioppo	Consumer	United States	New York Cr	New York	10024	East	OFF-BI-10000343									
3836	CA-2014-100363	08 April 2014	4/15/2014	Standard Class	JM-15655	Jim Mitchum	Corporate	United States	Glendale	Arizona	85301	West	OFF-FA-10000611									
3837	CA-2014-100363	08 April 2014	4/15/2014	Standard Class	JM-15655	Jim Mitchum	Corporate	United States	Glendale	Arizona	85301	West	OFF-PA-10004733									
9441	CA-2014-100391	25 May 2014	5/29/2014	Standard Class	BW-11065	Barry Weirich	Consumer	United States	New York Cr	New York	10035	East	OFF-PA-10001471									
6572	CA-2014-100678	18 April 2014	4/22/2014	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Houston	Texas	77095	Central	TEC-AC-10000474									
6569	CA-2014-100678	18 April 2014	4/22/2014	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Houston	Texas	77095	Central	OFF-AR-10001868									
6571	CA-2014-100678	18 April 2014	4/22/2014	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Houston	Texas	77095	Central	OFF-EN-10000056									
6570	CA-2014-100678	18 April 2014	4/22/2014	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Houston	Texas	77095	Central	FUR-CH-10002602									
9020	CA-2014-100706	16 December 2014	12/18/2014	Second Class	LE-16810	Laurel Elliston	Consumer	United States	Springfield	Virginia	22153	South	TEC-AC-10001314									
9021	CA-2014-100706	16 December 2014	12/18/2014	Second Class	LE-16810	Laurel Elliston	Consumer	United States	Springfield	Virginia	22153	South	FUR-FU-10002268									
6315	CA-2014-100762	24 November 2014	11/29/2014	Standard Class	NG-18355	Nat Gilpin	Corporate	United States	Jackson	Michigan	49201	Central	OFF-AR-10000380									
6316	CA-2014-100762	24 November 2014	11/29/2014	Standard Class	NG-18355	Nat Gilpin	Corporate	United States	Jackson	Michigan	49201	Central	OFF-LA-10003930									
6317	CA-2014-100762	24 November 2014	11/29/2014	Standard Class	NG-18355	Nat Gilpin	Corporate	United States	Jackson	Michigan	49201	Central	OFF-PA-10001815									
6318	CA-2014-100762	24 November 2014	11/29/2014	Standard Class	NG-18355	Nat Gilpin	Corporate	United States	Jackson	Michigan	49201	Central	OFF-PA-10004082									
9660	CA-2014-100860	26 March 2014	3/30/2014	Second Class	CS-12505	Cindy Stewart	Consumer	United States	Pomona	California	91767	West	TEC-AC-10001982									
9462	CA-2014-100867	19 October 2014	10/24/2014	Standard Class	BH-14125	Eugene Hildebrand	Home Office	United States	Lakewood	California	90712	West	TEC-PH-10004922									
8464	CA-2014-100881	28 March 2014	4/1/2014	Standard Class	DR-12940	Daniel Raglin	Home Office	United States	Albuquerque	New Mexico	87105	West	TEC-PH-10003273									
3888	CA-2014-100895	02 June 2014	6/6/2014	Standard Class	SV-20785	Stewart Voinsky	Consumer	United States	Roswell	Georgia	30076	South	OFF-AR-10004511									
3889	CA-2014-100895	02 June 2014	6/6/2014	Standard Class	SV-20785	Stewart Voinsky	Consumer	United States	Roswell	Georgia	30076	South	TEC-PH-10001425									
3890	CA-2014-100895	02 June 2014	6/6/2014	Standard Class	SV-20785	Stewart Voinsky	Consumer	United States	Roswell	Georgia	30076	South	OFF-ST-10001490									
7573	CA-2014-100916	21 October 2014	10/26/2014	Standard Class	FH-14275	Frank Hawley	Corporate	United States	Newport Ne	Virginia	23602	South	FUR-TA-10004607									
7572	CA-2014-100916	21 October 2014	10/26/2014	Standard Class	FH-14275	Frank Hawley	Corporate	United States	Newport Ne	Virginia	23602	South	OFF-ST-10003479									
7574	CA-2014-100916	21 October 2014	10/26/2014	Standard Class	FH-14275	Frank Hawley	Corporate	United States	Newport Ne	Virginia	23602	South	OFF-AR-10001022									

Let's transform this data to remove duplicate records step by step:

- Let's count each order id by using group by on order_id.

Order ID	Count
38	3
39	2
40	7
41	3
42	2
43	2
44	2
45	3
46	2
47	5
48	2
49	2
50	2
51	5
52	5
53	2
54	6
55	3
56	2
57	5
58	7
59	5
60	2
61	2
62	4

- In the below screenshot you can see all the selected columns like **"Order ID", "Order Date", "Customer ID", "Product ID", "Sales", "Quantity", "Discount", "Profit"** to find all the unique rows and in the filter you also see there is only one value that is 1 means there is not any duplicate rows.



Incorrect/Invalid Data Check

Check	Your result	Assessment
Missing values	0% empty, 0% errors	✓ Correct
Duplicate transactions	Count = 1	✓ Correct
Data types	Already appropriate	✓ Correct
Quantity	1–14, all > 0	✓ Correct
Sales	No values ≤ 0	✓ Correct
Discount	0–0.8	✓ Correct
Profit	Negative values exist	✓ Correct
Order Date ≤ Ship Date	All TRUE	✓ Correct

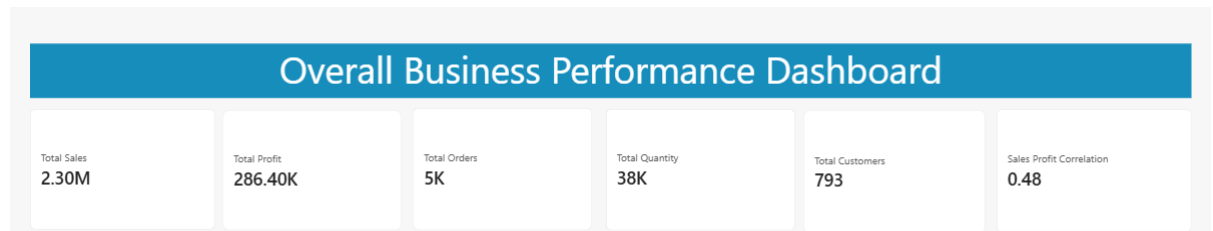
Task 4:

Solution:

Exploratory Data Analysis (EDA) is to understand the patterns, trends, relationships, and important characteristics present in the prepared dataset. Power BI was used to analyze sales, profit, orders, customers, categories, regions, segments, sub-categories, and products through different visualizations.

1. Overall Business Performance

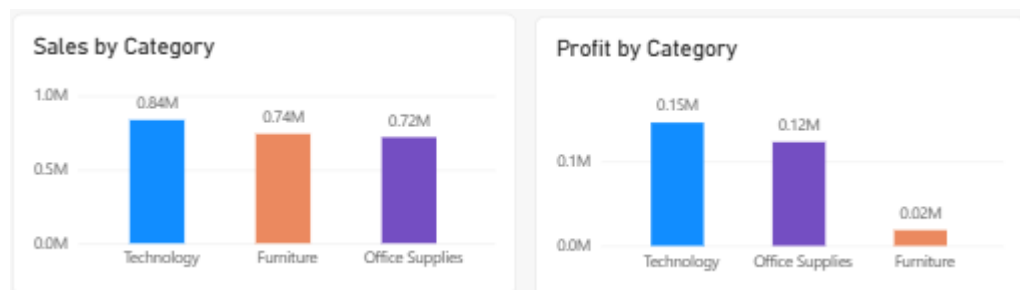
The overall performance of the dataset was analyzed using KPI cards as you can see in the below screenshot.



The dataset contains approximately **2.30 million in total sales** and **286.40K in total profit**. A total of approximately **5K orders** were analyzed, involving **793 customers** and **38K units of quantity**.

2. Category-wise Analysis

The **Sales by Category** and **Profit by Category** charts were used to compare the performance of different product categories.



Findings: -

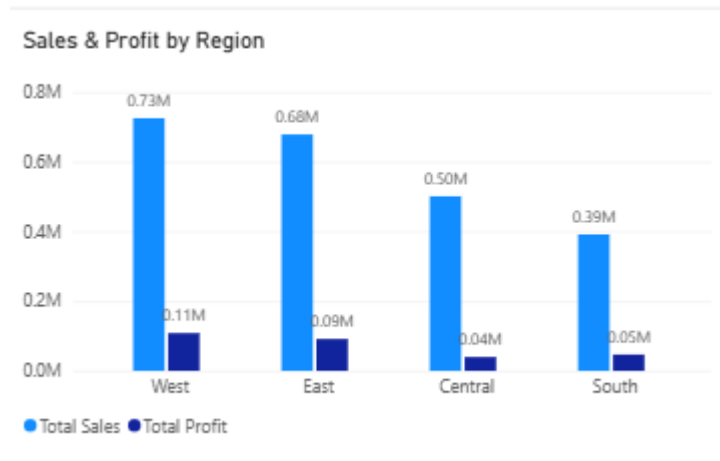
- **Technology** has the highest sales, approximately **0.84M**.
- **Furniture** generates approximately **0.74M** in sales.
- **Office Supplies** generates approximately **0.72M** in sales.
- Technology also generates the highest profit, approximately **145K**.
- Office Supplies generates approximately **122K** profit.
- Furniture generates only approximately **18K** profit despite having relatively high sales.

Conclusion: -

Technology is the strongest category in terms of both sales and profit. Furniture requires further investigation because its sales are relatively high but its profit contribution is considerably lower.

3. Regional Analysis

The **Sales & Profit by Region** chart was used to compare business performance across different regions.



Finding: -

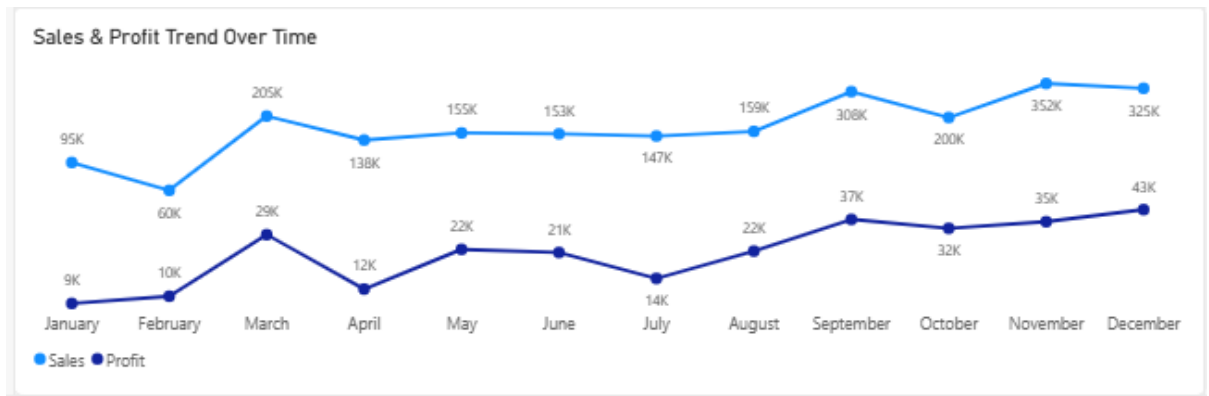
- **West** has the highest sales at approximately **0.73M**.
- **East** follows with approximately **0.68M**.
- **Central** generates approximately **0.50M**.
- **South** has the lowest sales at approximately **0.39M**.
- West also has the highest profit among the regions.

Conclusion: -

The West region is the strongest-performing region, while the South region has comparatively lower sales. This difference can be investigated further to understand regional customer demand and product performance.

4. Sales and Profit Trend Over Time

A **line chart** was used to analyze the monthly movement of sales and profit.



Finding: -

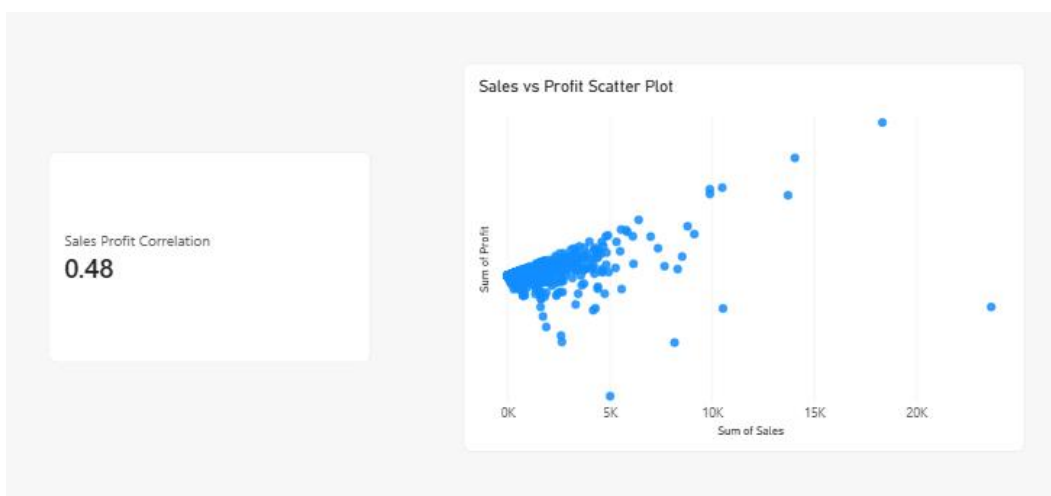
- Sales and profit fluctuate throughout the year.
- Sales increase significantly around **March** and again during **September–December**.
- The highest sales are observed around **November**, followed by December.
- Profit also shows variation throughout the year.
- The overall trend indicates that sales and profit do not remain constant throughout the year.

Conclusion: -

The business experiences variations in performance across different months. The higher sales toward the later part of the year may indicate stronger seasonal demand.

5. Sales and Profit Relationship

A **scatter plot** was used to examine the relationship between sales and profit.



Finding: -

The calculated **Sales-Profit Correlation is 0.48**.

Conclusion: -

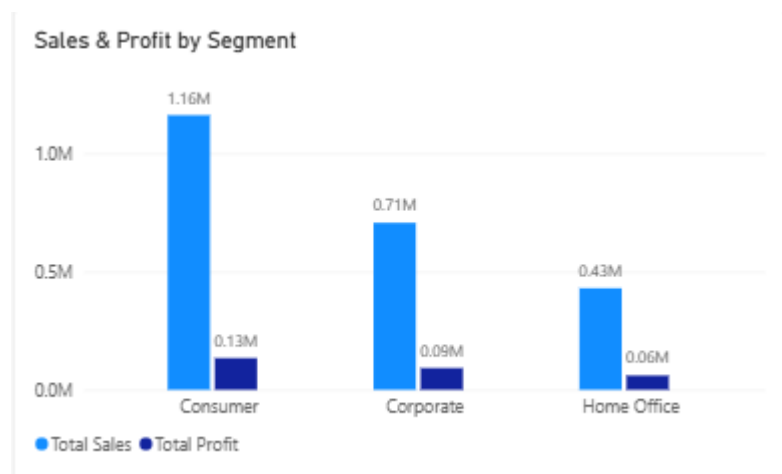
A correlation of **0.48** indicates a **moderate positive relationship** between sales and profit. In general, higher sales tend to be associated with higher profit, although the relationship is not perfect.

The scatter plot also shows some observations that are far away from the main group of points. These are potential outliers or high-value transactions.

The identified outliers were not removed because they may represent genuine business transactions rather than incorrect data.

6. Segment-wise Analysis

The **Sales & Profit by Segment** chart was used to understand the performance of different customer segments.



Finding: -

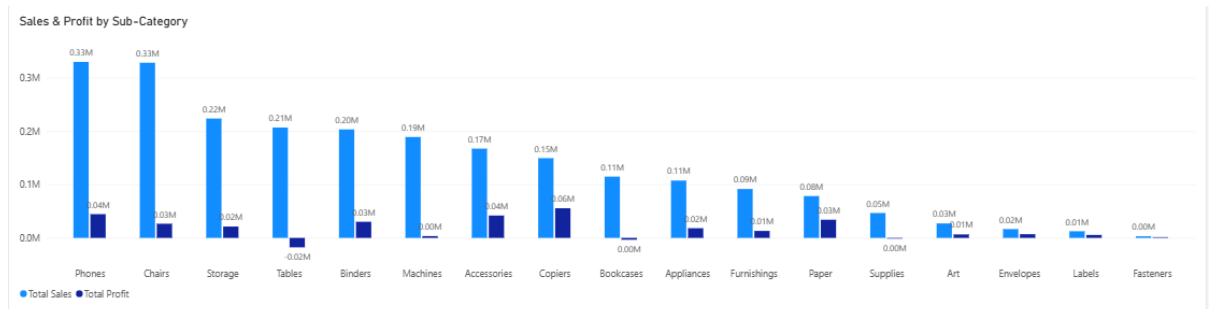
- **Consumer** is the highest-performing segment, with approximately **1.16M sales** and **0.13M profit**.
- **Corporate** generates approximately **0.71M sales** and **0.09M profit**.
- **Home Office** generates approximately **0.43M sales** and **0.06M profit**.

Conclusion: -

The Consumer segment contributes the largest portion of both sales and profit. Corporate is the second-largest contributor, while Home Office contributes the least among the three segments.

7. Sub-Category Analysis

The **Sales & Profit by Sub-Category** chart was used to examine detailed product-level categories.



Finding: -

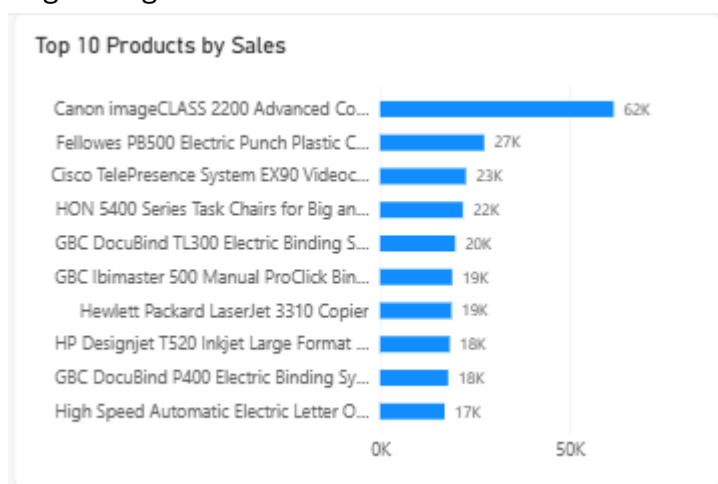
- **Phones and Chairs** are among the highest-selling sub-categories.
- Several sub-categories generate relatively low sales.
- Some sub-categories show **negative profit**.
- This indicates that high sales do not always guarantee high profitability.

Conclusion: -

The sub-category analysis highlights areas where sales performance and profitability differ. Sub-categories with negative or very low profit may require further investigation into discounts, pricing, product costs, or other business factors.

8. Top 10 Products by Sales

The **Top 10 Products by Sales** chart was created to identify the products generating the highest sales.



Finding: -

- **Canon imageCLASS 2200...** is the highest-selling product, with approximately **62K sales**.

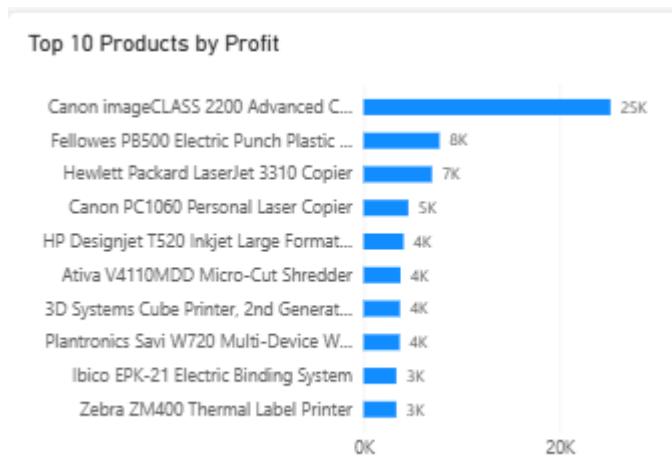
- Fellowes, Cisco, and other products also contribute significantly to sales.
- The chart helps identify the products that contribute most to total revenue.

Conclusion: -

A relatively small group of products contributes a significant amount of sales. These products can be considered important products for maintaining overall revenue.

9. Top 10 Products by Profit

The **Top 10 Products by Profit** chart was used to identify the most profitable products.



Finding: -

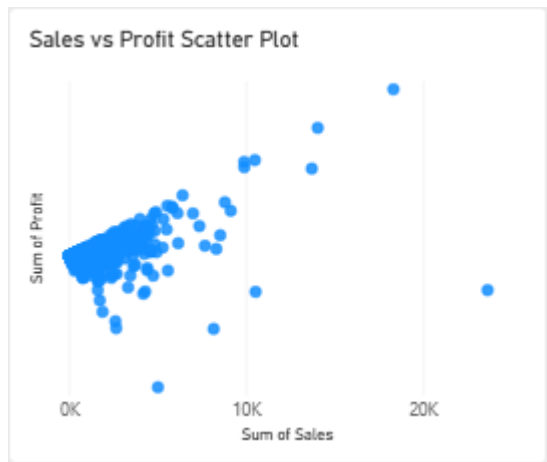
- **Canon imageCLASS 2200...** has the highest profit at approximately **25K**.
- Fellowes PB500 Electric... generates approximately **8K profit**.
- Hewlett Packard LaserJet... generates approximately **7K profit**.
- Other products in the top 10 generate approximately **3K–5K profit** each.

Conclusion: -

The top profitable products are not necessarily identical to the top-selling products. Therefore, both sales and profit should be considered when evaluating product performance.

10. Outlier Analysis

The scatter plot shows some observations that are significantly separated from the main concentration of data points. These observations can be considered potential outliers.



However, the outliers were **not removed** because there was no evidence that they represented incorrect records. They may represent genuine high-value orders or unusual business transactions.

Therefore, the outliers were retained for analysis to avoid removing potentially meaningful business information.

11. Key Insights from EDA

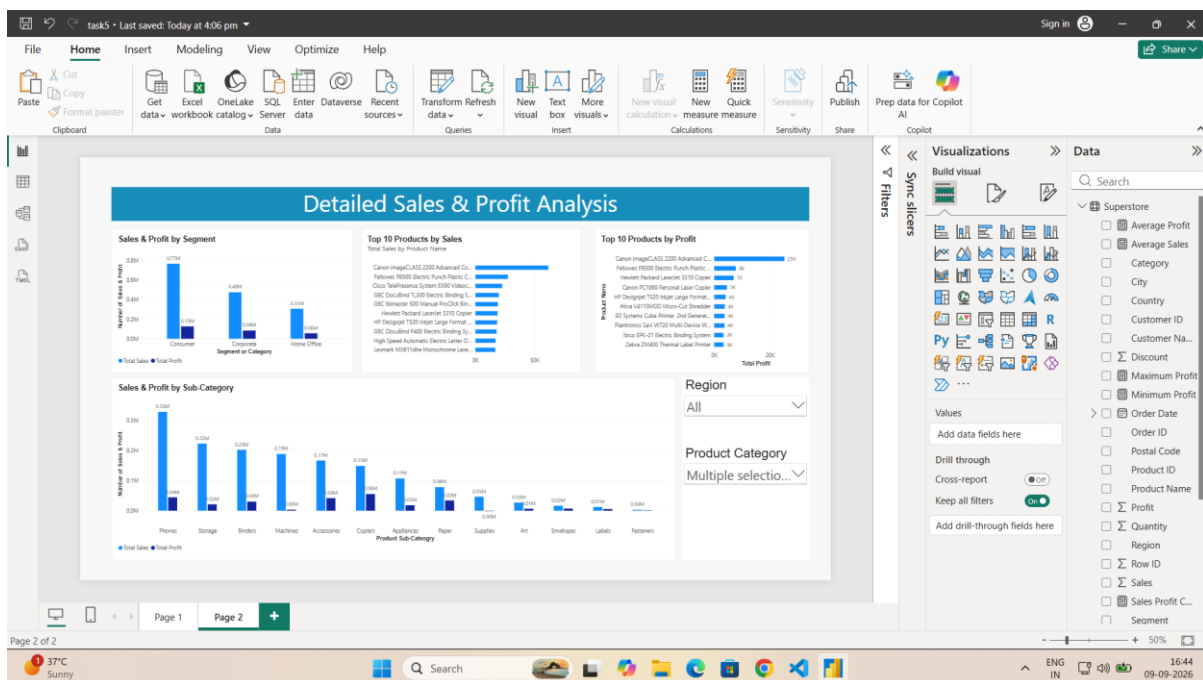
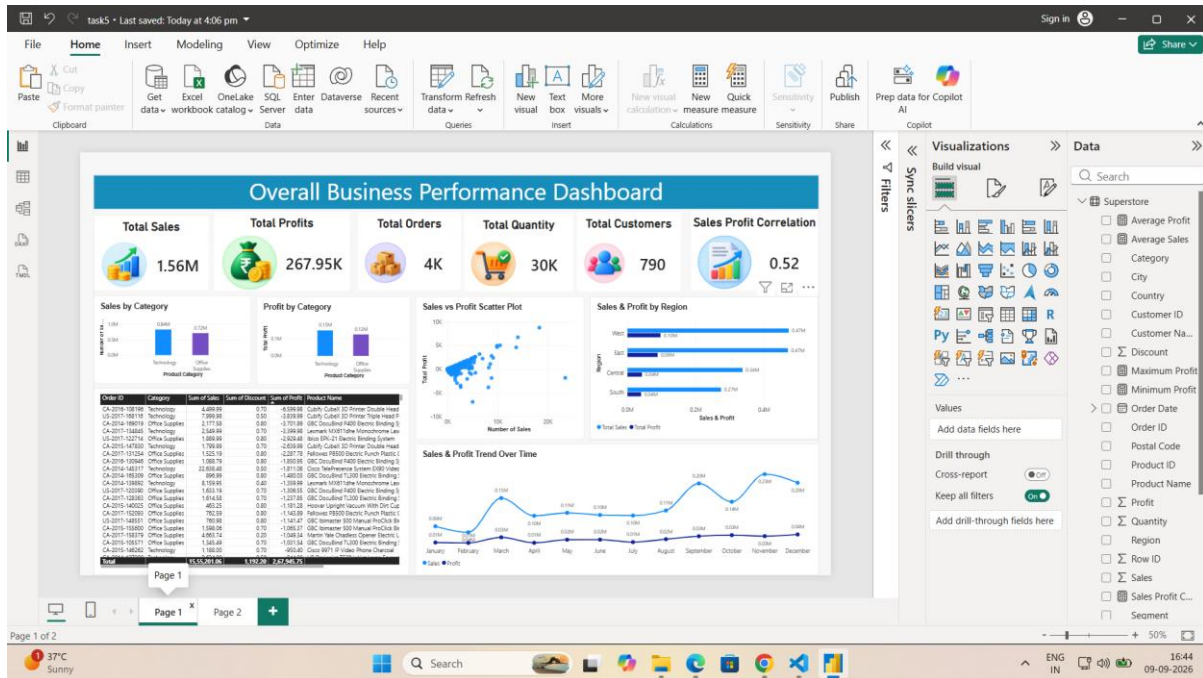
The major insights obtained from the analysis are:

1. **Technology** is the strongest category in terms of both sales and profit.
2. **Furniture** has relatively high sales but comparatively low profit.
3. **West** is the highest-performing region by sales.
4. **Consumer** is the strongest customer segment.
5. **Phones and Chairs** are among the highest-selling sub-categories.
6. Some sub-categories generate **negative profit**, indicating potential areas for improvement.
7. **Canon imageCLASS 2200...** appears among the leading products by both sales and profit.
8. Sales and profit have a **moderate positive correlation of 0.48**.
9. Some potential outliers were identified but retained because they may represent genuine transactions.
10. Sales and profit vary across months, with stronger performance visible toward the later part of the year.

Task 5:

Solution:

Created an interactive Power BI dashboard to analyze **sales and profit performance**. The dashboard includes KPI cards, category/segment analysis, regional analysis, product performance, time trends, and a sales-profit scatter plot. **Slicers for Region and Product Category** were added for interactive filtering.



Task 6:

Solution:

Developed an interactive two-page Power BI dashboard containing KPI cards, multiple charts, and slicers. The dashboard provides insights into sales, profit, orders, customers, categories, regions, segments, and products, with a clear and organized layout.

1. KPI Cards

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Total Customers
- Sales Profit Correlation

2. Visualizations – Page 1

- Sales by Category

- Profit by Category
- Sales vs Profit Scatter Plot
- Sales & Profit by Region
- Sales & Profit Trend Over Time
- Detailed Sales Table

3. Visualizations – Page 2

- Sales & Profit by Segment
- Top 10 Products by Sales
- Top 10 Products by Profit
- Sales & Profit by Sub-Category

4. Slicers / Filters

- Region
- Product Category

5. Dashboard Pages

- Page 1: **Overall Business Performance Dashboard**
- Page 2: **Detailed Sales & Profit Analysis**

6. Interactivity

Added slicers to allow users to filter and analyze the dashboard dynamically.

These are the screenshot you can see below:-

Page 1

Technology is the strongest category in terms of sales and profit, while Furniture has relatively high sales but lower profit.

2. Regional Performance

The West region has the highest sales and profit, while South has the lowest sales.

3. Customer Segment

Consumer is the largest and strongest customer segment, contributing the highest sales and profit.

4. Product Performance

A small number of products contribute significantly to sales and profit. The Top 10 product analysis helps identify these important products.

5. Sales vs Profit

Sales and profit have a moderate positive relationship. The correlation analysis shows that higher sales generally lead to higher profit, but high sales do not always guarantee high profitability.

- Business Recommendations

1. The company should investigate low-profit and negative-profit sub-categories and review their pricing, discounts, and costs.
2. The company should focus on high-performing products and the Consumer segment while developing strategies to improve sales in weaker regions and segments.

- Conclusion

The analysis shows that improving sales alone is not sufficient; the company should also focus on profitability, discounts, product performance, and regional performance to achieve sustainable business growth.

Task 8:

Solution:

The project was documented and organized for GitHub submission. A PowerPoint presentation was also created to communicate the data preparation, EDA, visualizations, Power BI dashboard, key insights, and business recommendations.

[Click here to view ppt](#)